GCI Global Final Project

Employee Attrition Analysis Using Logistic Regression

Overview of Employee Attrition

Employee attrition become major problem in every company. When a company experiences a high employee turnover rate, it can lead to several consequences both financially and psychologically. Financially, the organization must spend additional resources to recruit and train new employees. At the same time, existing employees may face increased mental strain, as they are often required to take on extra responsibilities until new hires are fully onboarded.

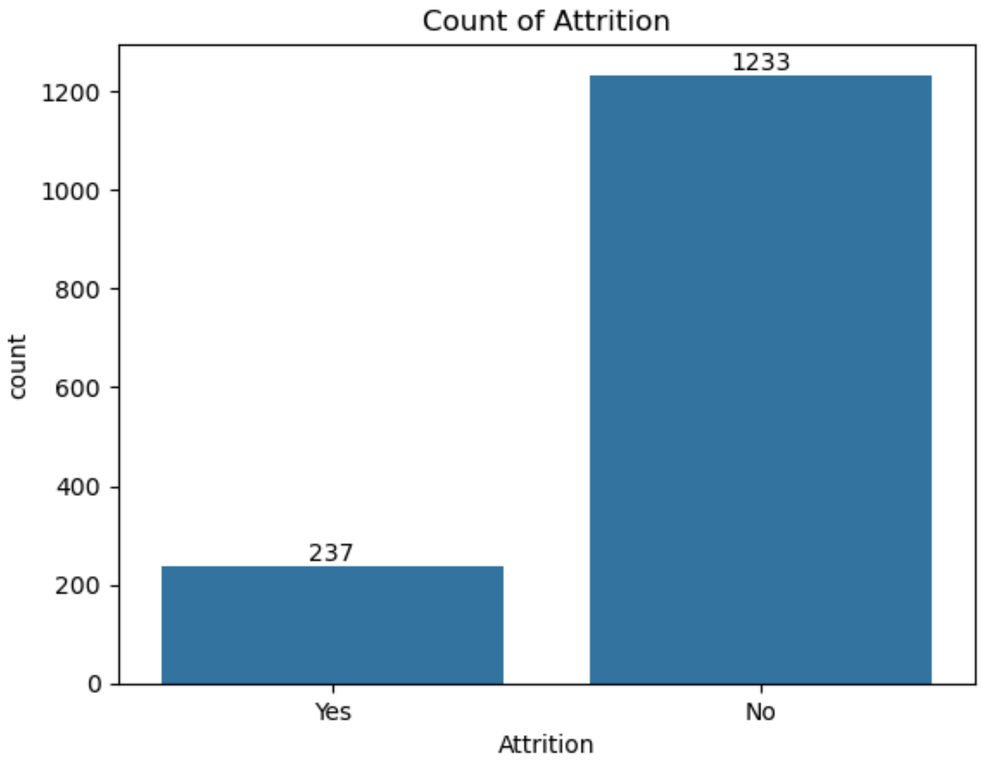
This analysis aims to identify:

1. **factors that cause attritions**

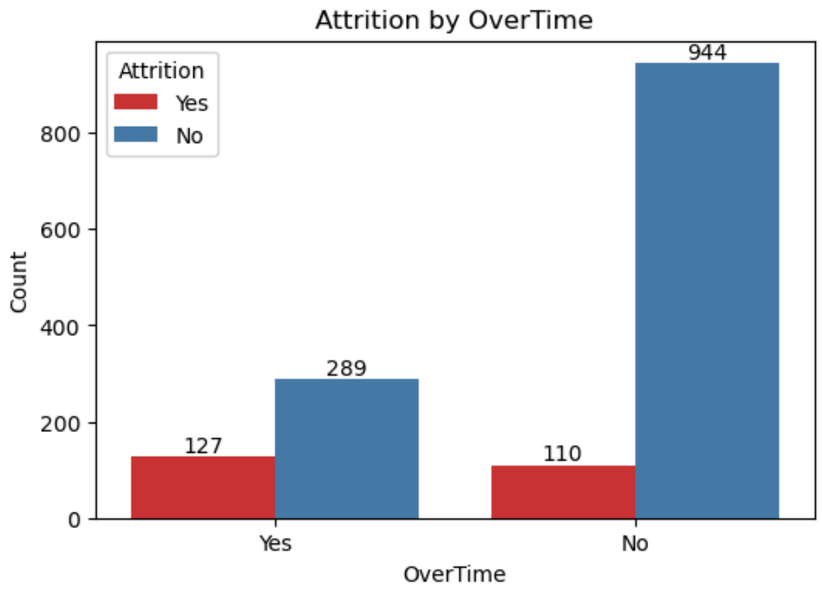
2. **To provide recommendations to reduce attrition rates Expertise**

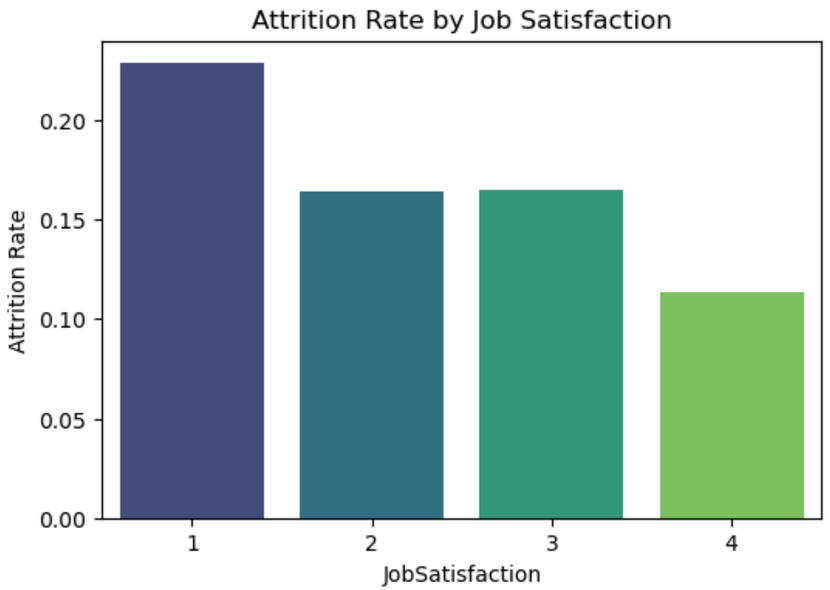
Dataset Overview

- dataset contains 1470 employees across the divisions and job roles
- target variable : Attrition (yes/no)
- Features : 35 attributes covers:
  - Demographic factors: Age, Gender, MaritalStatus
  - Job-related factors: JobRole, JobLevel, TotalWorkingYears, YearsAtCompany
  - Financial factors: MonthlyIncome, PercentSalaryHike
  - Workplace behavior: OverTime, JobSatisfaction, EnvironmentSatisfaction
- Class distribution is imbalanced with results **16% Attrition : yes** and **84% Attrition : No**



Exploratory Data Analysis

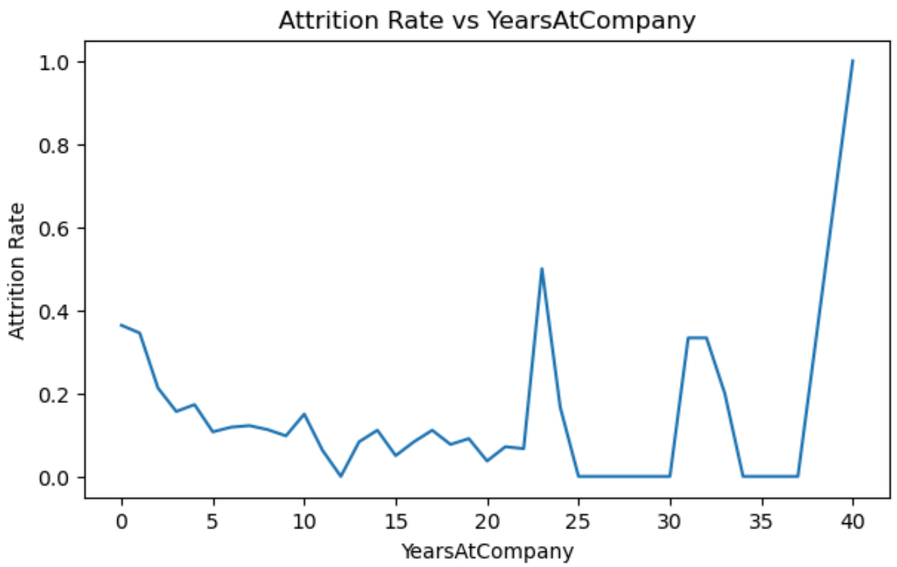


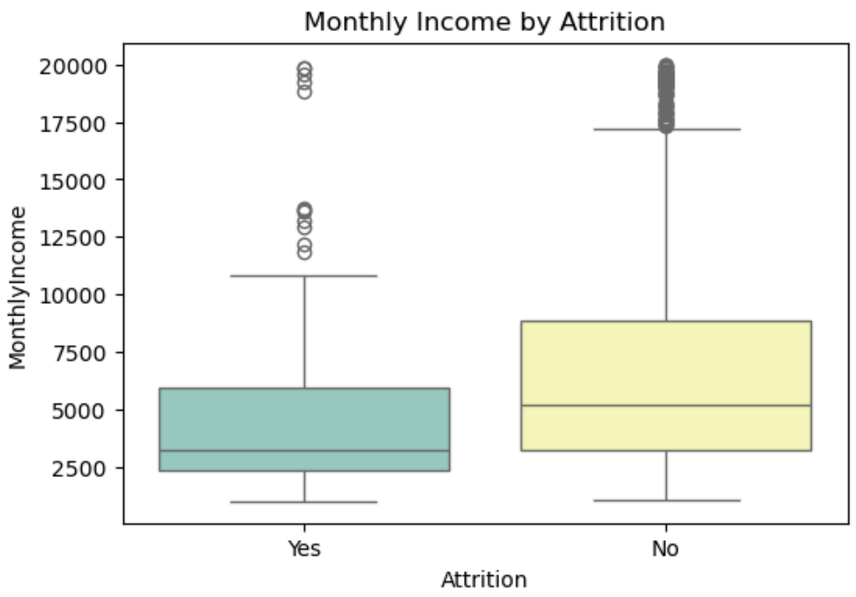


Overtime shows to have the strongest relationship with attritions. Employees who are overtime are nearly 3 times higher than those who are not overtime. this shows that overtime may reflects poor work life balance and work preassure. making it a significant factor to employee to turnover

Job satisfaction has a strong influence on employee attrition. Employees who receive a job satisfaction rating of ‘1’ are twice as likely to leave compared to those who receive a rating of ‘4’. This indicates that job satisfaction is an important factor in determining employee attrition rate

Exploratory Data Analysis



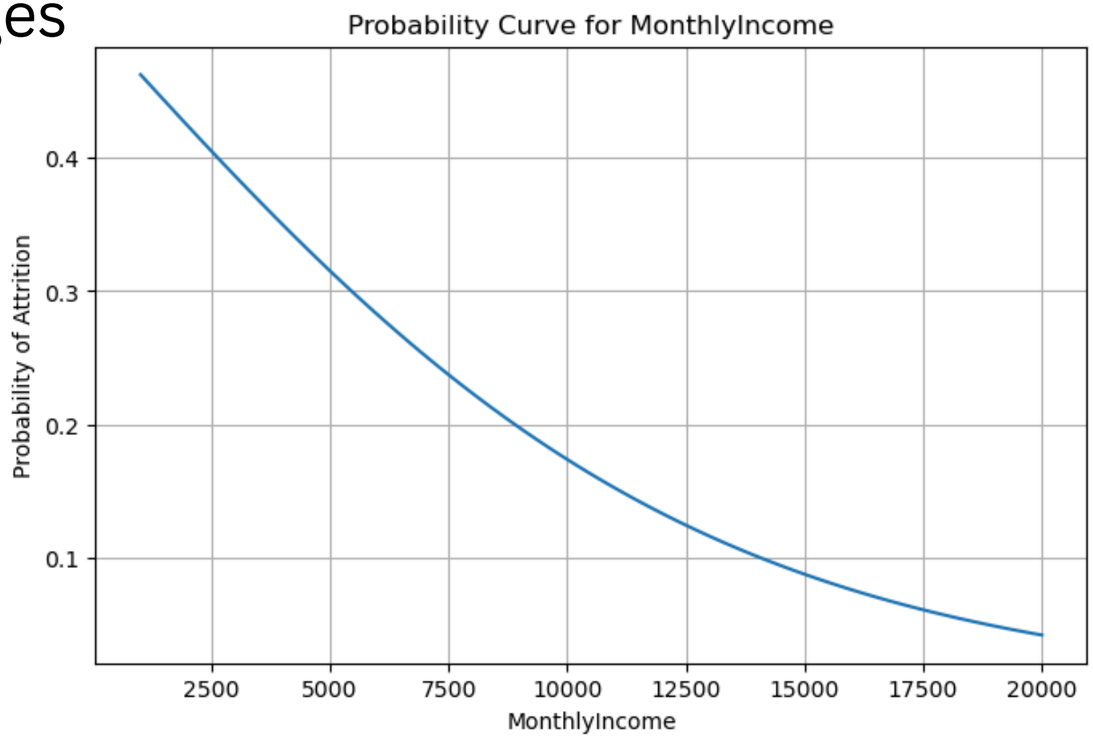


attrition rates at year 1 - 3 are quiet high, indicating early career employee are likely to leave. then attrition rates at year 4 to 10 are low, indicating strong retention after the initial adjustment period. A small spike appears again around 20+ years, possibly due to stagnation or retirement-related decisions. at last year 40 shows significant spikes indicating employees are retiring

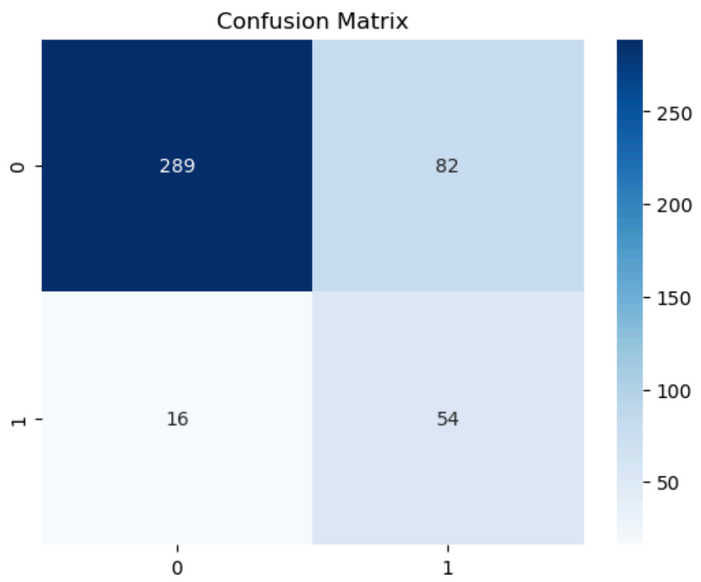
Employee with low monthly income are likely to resign due to financial dissatisfaction. mean while employees who have high monthly income are having low attrition rates. showing that monthly income is crucial for the employee since they who have lower income tend to leave.

Modeling Approach

- To analyze this data i use **Logistic Regression** Model
- Logistic Regression is used to predict the probability of specific outcome occurs (yes/no)
- if we change one factor Ex. “monthly_income” we can see how the model predicted attrition probability changes
- from this visualization we can see that the **lower the income the higher attrition rates.**



Model Performance



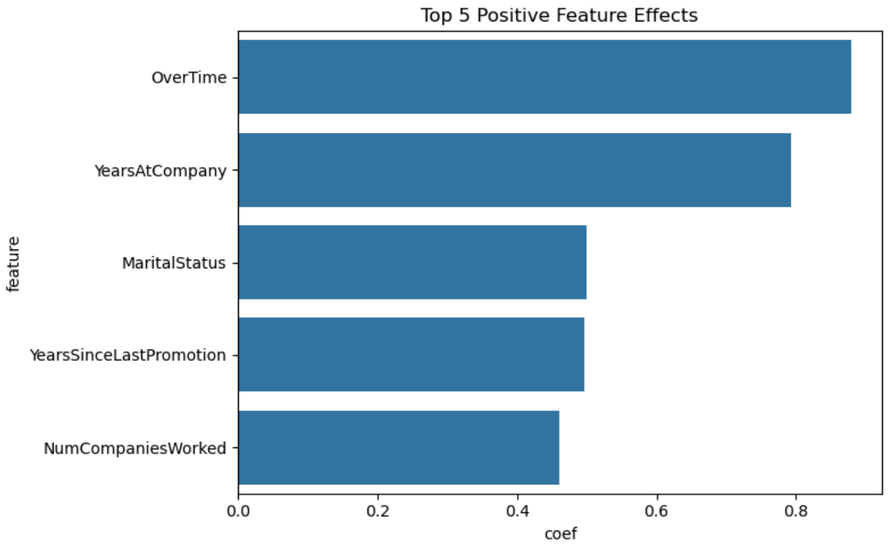
from 441 employee from the data test :

- 289 employees that are not resigning are predicted correctly
- 54 employees that are resigning are succesfully detected
- 82 employees are predicted to resign and actually not
- 16 employees are resigning but not detected

with this model we can conclude that :

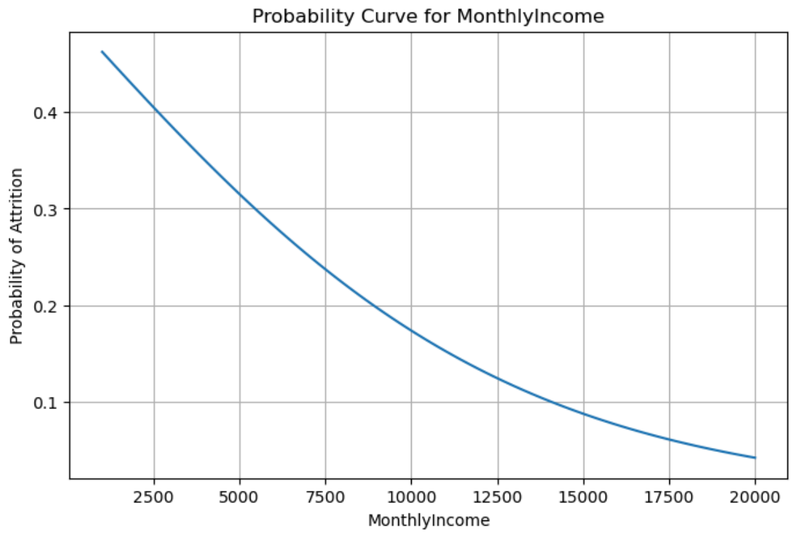
- this model can detect 77% employees that are really resigning
- this model has 78% accuration
- this model is effective to detect employees that have risk to resign besides that this model also has tedency to be alert → creates some false positive
- ROC-AUC of 0.84 indicates that the model can reliably distinguish between employees at risk of resigning and those who are not. It reflects strong pattern-recognition performance, though further evaluation is still needed for individual-level accuracy

Key Drivers of Attritions



these are the factors that most influence employee to resign
→ right bars increase the attrition rates
→ left bars decrease the attrition rates

- overtime is the main reason why many employee choose to resign
- followed by Years_At_company. this is reasonable because when once an employee has work for a long time they will retire eventually
- Employees who have previously worked at multiple companies tend to be more mobile and more likely to move again.
- Some departments are at higher risk of attrition than others.This could be due to differences in workload or team culture.



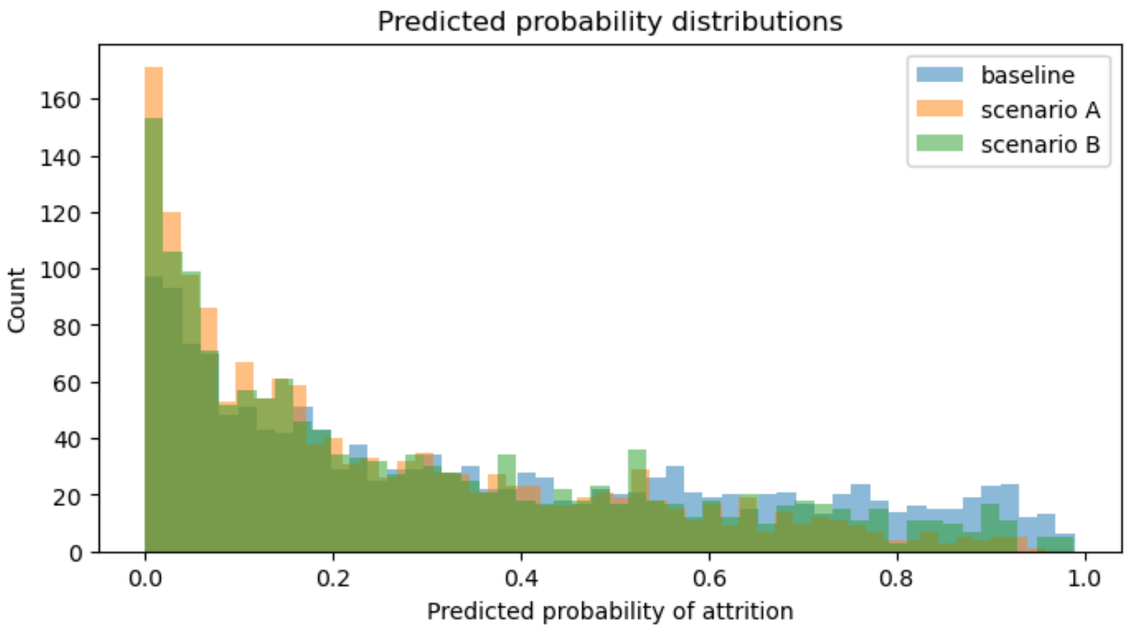
even though monthly income doesn’t enter the top 5 but monthly income has influence in attrition rates where
→ the more monthly income employee gets, the lower their attrition rates
→ the less monthly income employees get, the higher their attrition rates

Impact of Salary Increase and Overtime Reduction on Attrition Risk

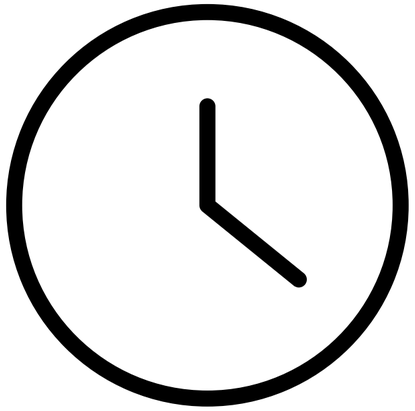
we make 2 prediction simulation :

1. first simulation we increase the monthly income to 20% then reduce their overtime to 0%
   - from the predicted attrition rates for the baseline we got 31%
   - once we start the first simulation we get that the predicted attrition reduce to 24%
2. the second simulation we increase the monthly income to 20% but reduce 50% of employees who currently do the overtime
   - when we increase the income to 20% and reduce employees who do the overtime to 50% we get predicted attrition reduce to 28%

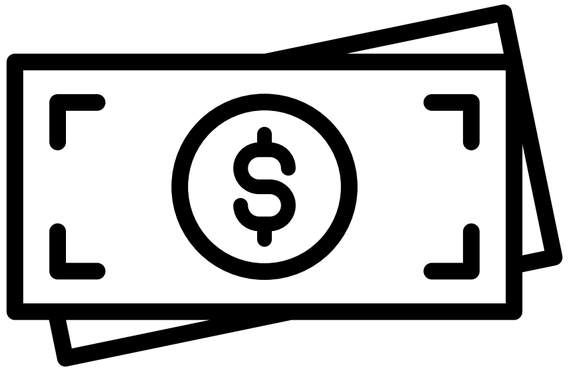
description	n	mean_prob	predicted_rate_percent
baseline (original)	1470	0.351144	31.088435
Income+20% & OverTime->0	1470	0.241928	16.666667
Income+20% & OverTime reduced by 50%	1470	0.283966	22.380952



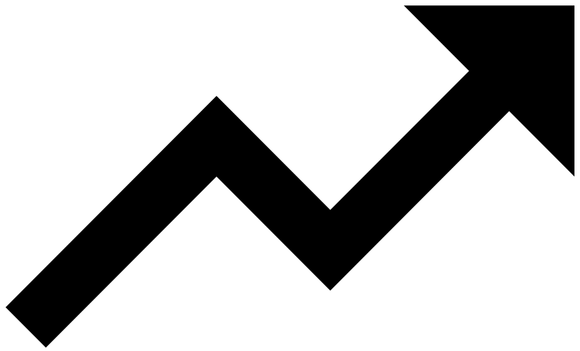
Recomendation to Reduce Attrition Rates



Set clear overtime limits per department to prevent excessive workload.



Provide fair compensation and reward high performers through bonuses or salary adjustments



Expand career advancement pathways to reduce career stagnation and improve retention